

Sohhom Bandyopadhyay

Applied AI / ML Engineer · PhD, Cognitive Science

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SUMMARY

Applied AI/ML engineer with a PhD in Cognitive Science (IIT Gandhinagar) and 3+ years of industry experience shipping with LLMs, classical ML, and signal processing. I move comfortably across the stack: scoping an ill-defined problem, designing the evaluation, building the prototype, and saying plainly when a method will not work. Recent work is in quantitative finance — trader-assistance tools, options analytics, and agentic LLM systems — built end to end. Background strength in measurement: I am most useful where the hard part is figuring out *what to measure* and whether a result is real.

EXPERIENCE

Applied AI Engineer — Trading Systems

Punch · Remote · *October 2024 – May 2026*

- Built an LLM-powered trading copilot that read live chart context and answered interactive market questions for experienced traders; designed the prompt structure, tool schema over market-data APIs, and the evaluation harness.
- Shipped rule-based technical-pattern recognition (head-and-shoulders, double tops/bottoms, trendline and support/resistance breaks) over OHLC data, surfaced as live chart annotations.
- Delivered options-analytics dashboards computing Greeks (delta, gamma, theta, vega) and max-pain across expiries; Python back end integrated with the firm's front end.
- Prototyped agentic trade-recommendation flows; studied market microstructure (market making, order flow, venue fragmentation) to ground the tooling in how prices actually form.
- Sole ML contributor on the engagement; worked directly with product and engineering leads, including pushing back on approaches that would not have held up.

Independent Researcher / Builder — Quantitative Trading

Self-directed · Remote · *July 2024 – Present*

- Designing an end-to-end automated trading system in Python: data ingestion, feature engineering, signal generation, backtesting, and execution logic, on Indian (NSE/BSE) market data.
- Built reusable components for transaction-cost modelling, strategy evaluation, and risk metrics; treat the cost-and-survivorship “break-even test” as the first filter on any idea.
- Stack: Python, pandas, NumPy, scipy, scikit-learn, Polars, broker APIs.

Artificial Intelligence Researcher

Playpower Labs · Remote (India) · *September 2023 – June 2024*

- Senior individual contributor reporting to the CEO; embedded with product and engineering as needed.
- Studied alignment of LLMs with human emotion using an emotion-theory-grounded approach; this work became a co-authored paper in *Advanced Design Research* (2024).
- Scoped and evaluated LLM and diffusion-model applications for EdTech, including automated essay scoring; produced feasibility analyses that informed build/buy/defer decisions.

R&D Researcher — Human–AI Interaction

Playpower Labs · Remote (India) · *July 2022 – April 2023*

- Trained and evaluated early Stable Diffusion image-generation pipelines and ran a literature review on human-preference elicitation for AI alignment, used internally to frame product directions.

Graduate Teaching Fellow

IIT Gandhinagar · *January – April 2019*

- Co-taught the MSc Cognitive Science core course *Computation and Cognition* (CG 501) with Prof. Krishna Prasad Miyapuram. Independently ran 17 classes, designed quizzes, a project, and the end-semester exam, and built fill-in-the-blank Jupyter assignments covering perceptron/MLP, linear regression, k-NN, and game theory.

EDUCATION

PhD, Cognitive Science

IIT Gandhinagar · *2017 – 2022* · *CPI 8.68*

Thesis: *Psychoacoustic assessment of active-noise-control (ANC) headphones*. Developed black-box protocols to measure what ANC does to the desired playback signal — not just how much noise it cancels — across four commercial headphones. Combined signal processing, a computational auditory-nerve model, human listening experiments, and ten instrumental speech-quality metrics. An external examiner described the methodological rigour as “excellent.” Full thesis

MSc, Cognitive Science

IIT Gandhinagar · *2015 – 2017* · *CPI 8.85*

Thesis: *Functional Link Artificial Neural Network for keystroke-dynamics authentication*. Implemented a FLANN classifier from scratch in NumPy and benchmarked it against MLP, random forest, k-NN, and decision trees; it gave the best AUROC and equal-error-rate on free-text keystroke data.

BSc (Hons), Computer Science

Scottish Church College, University of Calcutta · 2012 – 2015 · 73.7% (*First Class*)

PUBLICATIONS

- Lomas, J. D., van der Maden, W., **Bandyopadhyay, S.**, et al. (2024). Evaluating the alignment of AI with human emotions. *Advanced Design Research*, 2(2), 88–97.
- Mukhopadhyay, D., **Bandyopadhyay, S.**, Venkatesh, E., & Menon, S. (2024). Complexity-entropy, eye-tracking and emotional responses to 40 paintings of Raja Ravi Varma. *International Journal of Arts and Technology*, 15(2), 170–207.
- Fonseca, A. F., **Bandyopadhyay, S.**, Louçã, J., & Manjaly, J. A. (2019). Caste in the news: a computational analysis of Indian newspapers. *Social Media + Society*, 5(4).

SELECTED PROJECTS

Quantitative investigations of NSE/BSE markets

Ongoing, 2024–

A running investigation of Indian equity and derivatives markets: transaction-cost and break-even analysis, intraday microstructure, and options-volatility studies. Write-ups and tooling at [the project site].

Keystroke-based user authentication

MSc thesis, 2016–2017

Neural network built from scratch in NumPy to identify users from typing patterns (dwell and flight times, digraph features); proof-of-concept Flask login portal. Explored what keystroke dynamics reveal about the chunking of motor sequences.

Natural-language search over an academic database

Course project, 2018

Team-built natural-language query interface over 500k+ research papers. Designed the ingestion pipeline and schema; coordinated team responsibilities and the tech-stack decisions.

TECHNICAL SKILLS

Languages: Python, R, MATLAB, JavaScript, SQL, C++, Bash

ML / data: scikit-learn, NumPy, pandas, Polars, scipy, statsmodels, matplotlib; statistical modelling and experiment design in R

LLMs / GenAI: prompt design, tool use and agentic flows, evaluation harnesses, Hugging Face Transformers, CLIP, Stable Diffusion / diffusion models

Signal processing: scipy.signal, MATLAB, psychoacoustic and auditory-model toolboxes

Web / deployment: Flask, Node.js, MongoDB, Netlify, Vercel, Git

Domains: quantitative finance (options Greeks, technical analysis, backtesting, microstructure), psychoacoustics, applied ML, HCI, EdTech